



Lube Oil Flushing Checklist

ISO-compliant flushing procedure checklist covering mobile centrifuge / flushing rig setup, oil sampling protocol, cleanliness verification, and final sign-off.

Reference standards: ISO 4406:2021 (fluid cleanliness classification by particle count) and ISO 16889 (multi-pass filter test / Beta ratio rating used to select flushing filter elements). This checklist summarizes common industry practice under these standards; it does not reproduce their full text.

Customer / Site:		Equipment:	
Oil Grade (ISO VG):		Tank Capacity (L):	
Target Cleanliness Class:	ISO 4406 ___ / ___ / ___	Date:	

1. Pre-Flushing Setup

#	Checkpoint	Spec / Reference	Done	Remarks
1	Drain and inspect reservoir for sludge, water, varnish	Visual inspection	☐	
2	Confirm flushing rig pump capacity and filter micron rating	3-10x normal system flow	☐	
3	Install temporary jumper hoses bypassing bearings/servos	Protects sensitive components	☐	
4	Fit coarse strainer at rig inlet to catch loose debris	100-150 micron	☐	
5	Verify oil grade compatibility with system (no mixing grades)	Match ISO VG / additive type	☐	

2. Flushing Rig Operation

#	Checkpoint	Spec / Reference	Done	Remarks
1	Circulate oil through rig without heater initially (cold flush)	Dislodge loose particles	☐	
2	Ramp oil temperature to 55-65 deg C for varnish/deposit removal	Do not exceed OEM max temp	☐	
3	Induce turbulent flow — vary flow rate / tap piping periodically	Reynolds no. > 4000 preferred	☐	
4	Rotate through all branch lines and dead-legs in sequence	No line left unflushed	☐	
5	Change/inspect flushing filter elements as differential pressure rises	Log each element change	☐	

3. Oil Sampling & Cleanliness Verification

#	Checkpoint	Spec / Reference	Done	Remarks
1	Draw sample after minimum 4 hours of continuous flushing	From live, turbulent point	☐	



#	Checkpoint	Spec / Reference	Done	Remarks
2	Send sample for particle count (ISO 4406) / moisture (ppm)	Accredited lab or portable counter	☐	
3	Compare result against target cleanliness class	See target above	☐	
4	Repeat flushing cycle if target class not met	Re-sample after each cycle	☐	
5	Record final particle count certificate number	Attach lab report	☐	

4. Reinstatement & Sign-Off

#	Checkpoint	Spec / Reference	Done	Remarks
1	Remove jumper hoses; reconnect bearings, servos, coolers	Torque per OEM spec	☐	
2	Replace final in-line filter elements with new OEM elements	Do not reuse flushing elements	☐	
3	Top up reservoir to correct level with same oil batch	Check dipstick / sight glass	☐	
4	Run system on normal circulation, re-check for leaks	30 min minimum run	☐	
5	Final cleanliness re-verified post-reinstatement	Confirms no re-contamination	☐	

5. ISO 4406 Cleanliness Code Reference

The ISO 4406 code reports particle counts per millilitre of fluid at three sizes, written as three numbers (e.g. 18/16/13) for particles greater than or equal to 4, 6, and 14 microns respectively. Each step up the scale represents a doubling of the particle count for that size — the scale is logarithmic, not linear.

Scale Number	Particles / mL: More Than	Up To and Including
24	80,000	160,000
23	40,000	80,000
22	20,000	40,000
21	10,000	20,000
20	5,000	10,000
19	2,500	5,000
18	1,300	2,500
17	640	1,300
16	320	640
15	160	320
14	80	160
13	40	80



Scale Number	Particles / mL: More Than	Up To and Including
12	20	40
11	10	20

6. Typical Target Cleanliness by Component Sensitivity

System / Component	Typical Target ISO 4406 Code
Steam turbine journal / tilting-pad bearings	18/16/13
Electro-hydraulic control (EHC) / servo valve systems	16/14/11
General industrial gearboxes	19/17/14
Standard hydraulic pumps and directional valves	18/16/13
High-precision proportional / servo valves	16/14/11 or cleaner

These are general industry starting points, not universal requirements — always confirm the target class against the OEM manual or the most contamination-sensitive component on the system (e.g. a servo valve tightens the target for the whole circuit it shares). Flushing filter elements should carry a Beta ratio of 200 or higher (per ISO 16889) at the target micron size to reliably achieve and hold the target class.

Sign-Off

Engineer (Keshav Enterprises):		Date:	
Customer Representative:		Date:	

This checklist reflects general good practice for turbine lube oil system flushing. Target cleanliness class, flushing duration, and temperature limits should always be confirmed against the OEM manual and the oil manufacturer's data sheet for the specific system.